

Cats vs Dogs - End-to-End MLOps Pipeline

MLOps (AIMLCZG523) - Assignment 2

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Repository: https://github.com/MadihaFathima/bits_mlops_assignment_2

Use case: Binary image classification (Cats vs Dogs) for a pet adoption platform

Objective

Design and implement an end-to-end MLOps pipeline for model building, artifact/image creation, packaging, containerization, and CI/CD-based deployment using open-source tools.

Dataset

Cats and Dogs binary classification dataset - specifically the **Microsoft Cats vs Dogs dataset** (shaunthesheep/microsoft-catsvsdogs-dataset on Kaggle), the original dataset behind the classic Kaggle "Dogs vs. Cats" competition. Chosen because it is raw and unsplit, matching the assignment's requirement to preprocess and split the data. 25,000 images total (12,500 per class); a balanced 4,000-image subsample (2,000/class) was used for training given CPU-only compute, split 80/10/10 into train/validation/test. All images preprocessed to 224×224 RGB. Training-time data augmentation (random horizontal flip, random rotation) applied on the training split only.

Every claim in this report reflects behavior that was actually run and verified against the live system - not planned or expected behavior.

M1: Model Development & Experiment Tracking 10M

Objective: Build a baseline model, track experiments, and version all artifacts.

1. Data & Code Versioning

Git for source code versioning; DVC for dataset versioning (local remote at `D:/Bits-SEM3/dvc-storage/mlops_assignment_2`).

Both the raw dataset (25,000 images, `data/raw/PetImages.dvc`) and the preprocessed dataset (4,000 images, `data/processed.dvc`) are tracked as separate DVC-versioned artifacts. The raw dataset contains a small number of corrupt/zero-byte files (a known quirk of this dataset's original release), filtered out during preprocessing via `is_valid_image()`.

```
(venv) D:\Bits-SEM3\mlops_assignment_2>git log --oneline
7e94159 (HEAD -> main, origin/main, origin/HEAD) Add README with setup and usage instructions
1641bcc Revert smoke test port; add post-deployment eval script and M4 negative-test evidence
9615cfb TEMP: deliberately break smoke test port to verify pipeline fails closed
f84ca1b Add request logging and Prometheus metrics for M5 monitoring
fe444e1 Increase rollout status timeout to 600s for slow multi-replica startup
9b2d1cf Switch deploy job to PowerShell to avoid Windows bash ambiguity entirely
2c8ebf0 Fix deploy job: pin shell to Git Bash explicitly, not ambiguous WSL shim
4420cad Fix service.yaml: correct port to 8080, was still committed as 80
90f2ede Add automated deploy job to CI/CD pipeline with post-deploy smoke test
1f3300b Fix pod startup: add startupProbe, PYTHONUNBUFFERED for slow torch import under CPU limits
03bd524 Fix CI: run pytest via python -m so src package resolves
bdf5c8e Add GitHub Actions CI pipeline: test, build, smoke-test, push to GHCR
f816c7d Add unit tests for preprocessing and inference
c691a71 Add FastAPI inference service and Dockerfile for M2
a6c6b68 Document MLflow filestore troubleshooting; gitignore stray mlflow.db
9f56713 Add baseline CNN, training script, trained model and MLflow-tracked run artifacts
d3d370a Add preprocessing script; DVC-track processed dataset
135d28c Add RUNBOOK documenting setup steps so far
d4b5ba8 Add dataset license and readme from Kaggle source
5256604 Track raw Cats vs Dogs dataset with DVC
86108ba Initialize DVC with local storage remote

(venv) D:\Bits-SEM3\mlops_assignment_2>git log --oneline -- data/raw/PetImages.dvc data/processed.dvc
d3d370a Add preprocessing script; DVC-track processed dataset
5256604 Track raw Cats vs Dogs dataset with DVC
```

2. Model Building

Baseline model: SimpleCNN (PyTorch) - 3 conv+ReLU+maxpool blocks (3→32→64→128 channels), adaptive average pooling, small fully-connected head. Saved as `models/cnn_baseline.pt` (`torch.save(state_dict)`).

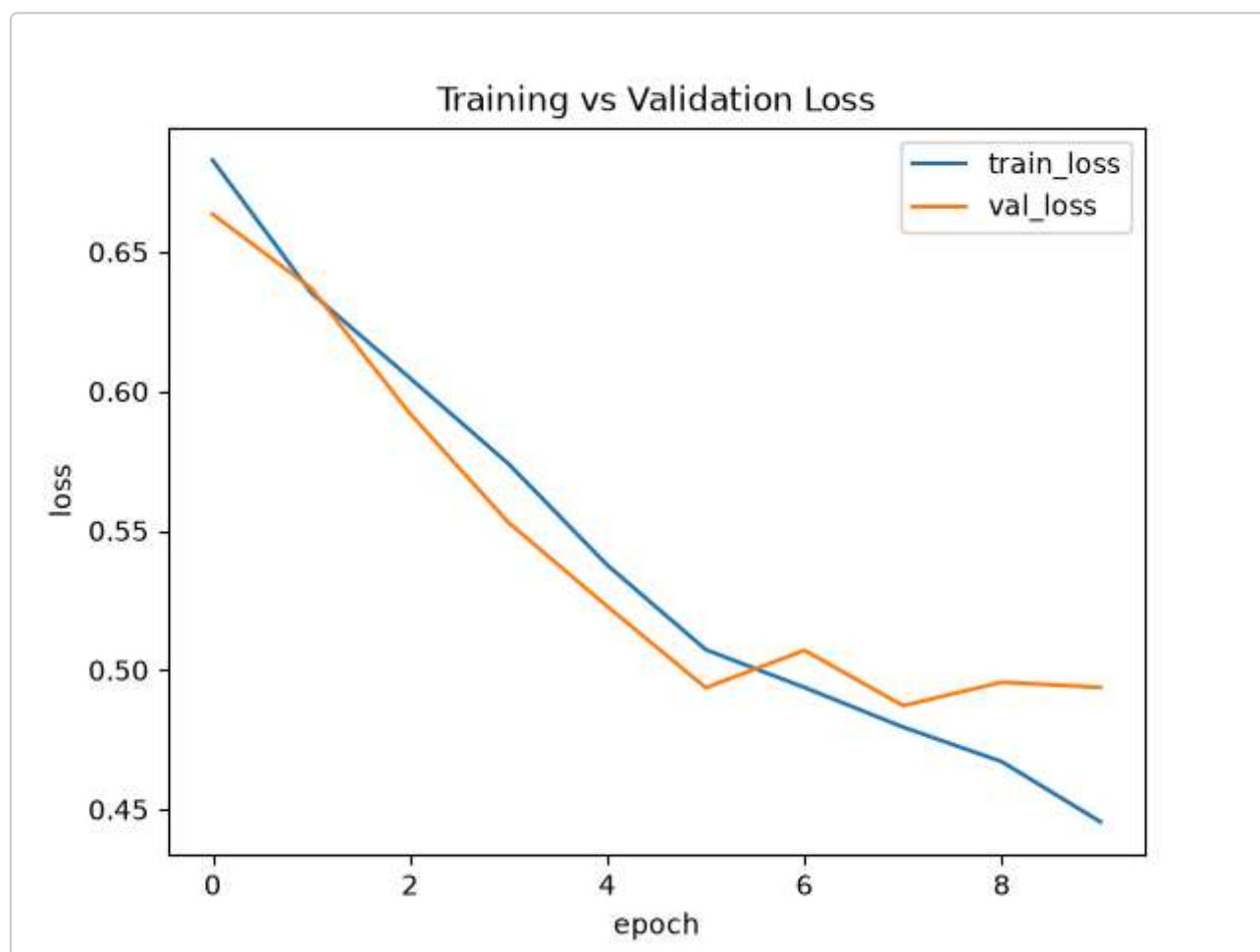
Deliberately a shallow architecture given CPU-only training. Trained for 10 epochs with Adam, BCEWithLogitsLoss, on the 4,000-image balanced subsample (3,200 train / 400 val / 400 test).

3. Experiment Tracking

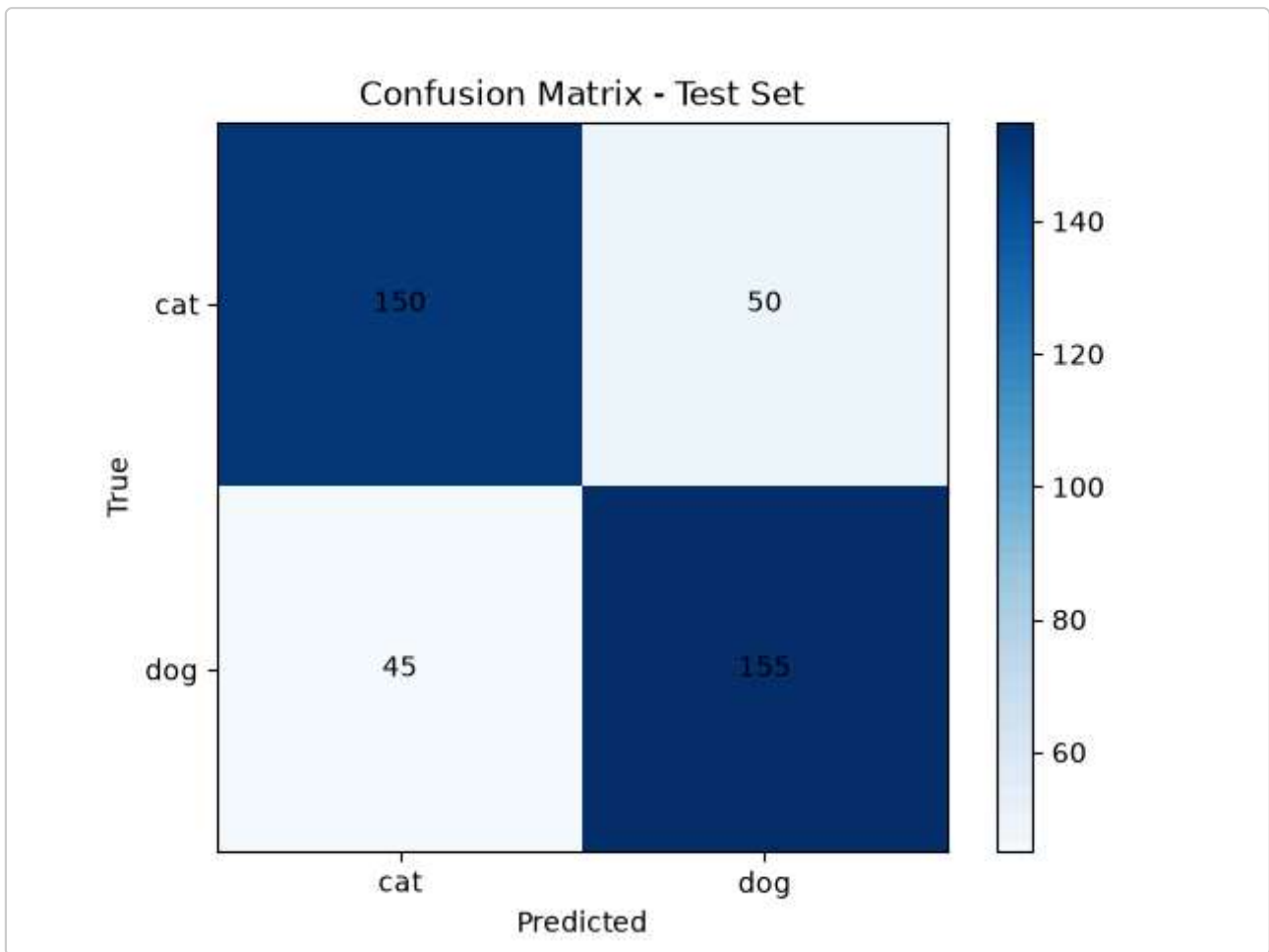
MLflow, experiment `cats-vs-dogs-classification` . Logs parameters, per-epoch metrics, final test metrics, confusion matrix, loss curve, and the trained model.

Metric	Value
Test accuracy	0.7625
Test precision	0.7561
Test recall	0.7750
Test F1	0.7654

Loss curve (train vs. validation, per epoch):



Confusion matrix (test set):



Starting the MLflow UI (note: this MLflow version's real tracking metadata lives in `mlflow.db`, not the `mlruns/` folder - see `RUNBOOK.md` for why):

```

C:\Windows\System32\cmd.exe x + v
PS D:\Bits-SEM3\mlops_assignment_2> venv\Scripts\python.exe -m mlflow ui --backend-store-uri sqlite:///mlflow.db
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.0 and configure --allow
ed-hosts and --cors-allowed-origins.
2026/08/30 14:33:46 WARNING mlflow.server: MLflow job execution requirements not met (MLflow job backend does not support Windows system.). Server will star
t without job execution support. Errors will be surfaced at job invocation time.
2026/08/30 14:33:47 INFO: Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/08/30 14:33:47 INFO: Started parent process [2120]
D:\Bits-SEM3\mlops_assignment_2\venv\Lib\site-packages\mlflow\server\fastapi_app.py:19: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated
and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_envi
D:\Bits-SEM3\mlops_assignment_2\venv\Lib\site-packages\mlflow\server\fastapi_app.py:19: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated
and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_envi
D:\Bits-SEM3\mlops_assignment_2\venv\Lib\site-packages\mlflow\server\fastapi_app.py:19: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated
and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_envi
D:\Bits-SEM3\mlops_assignment_2\venv\Lib\site-packages\mlflow\server\fastapi_app.py:19: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated
and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
  from starlette.middleware.wsgi import WSGIResponder, build_envi
2026/08/30 14:33:57 INFO: Started server process [6772]
2026/08/30 14:33:57 INFO: Started server process [22872]
2026/08/30 14:33:57 INFO: Waiting for application startup.
2026/08/30 14:33:57 INFO: Waiting for application startup.
2026/08/30 14:33:57 INFO: Started server process [22680]
2026/08/30 14:33:57 INFO: Waiting for application startup.
2026/08/30 14:33:57 INFO: Started server process [23140]
2026/08/30 14:33:57 INFO: Waiting for application startup.
2026/08/30 14:33:57 INFO: Application startup complete.
2026/08/30 14:33:57 INFO: Application startup complete.
2026/08/30 14:33:57 INFO: Application startup complete.
2026/08/30 14:33:57 INFO: Application startup complete.

```

Run overview - logged parameters and metrics for the successful run:

mlflow 3.15.1

GenAI Model training

cats-vs-dogs-cl...

Overview Training runs Observability Traces Sessions Evaluation Judges Review Datasets Evaluation runs Prompts & versions Playground

cats-vs-dogs-classification > Runs > puzzled-colt-687

Overview Model metrics System metrics Traces Artifacts

Description No description

Metrics (8)

Metric	Value	Models
train_loss	0.44554504692554475	-
train_accuracy	0.7984375	-
val_loss	0.4937513518333435	-
val_accuracy	0.7575	-
test_accuracy	0.7625	model
test_precision	0.7560975609756098	model
test_recall	0.775	model

About this run

Created at 08/23/2026, 09:11:49 AM

Created by Madiha Fathima

Experiment ID 1

Status Finished

Run ID 9a622c5b19bb4e8dab8280c7fb374abf

Duration 20.5min

Source D:\Bits-SEM3\mlops_assignment_2\src\models\train.py

Registered prompts —

Datasets None

Artifacts tab - loss curve and confusion matrix logged and viewable inline, alongside the serialized model and metadata:

mlflow 3.15.1

GenAI Model training

cats-vs-dogs-cl...

Overview Training runs Observability Traces Sessions Evaluation Judges Review Datasets Evaluation runs Prompts & versions Playground Prompts Agent versions AI Gateway New Docs

cats-vs-dogs-classification > Runs > puzzled-colt-687

Overview Model metrics System metrics Traces Artifacts

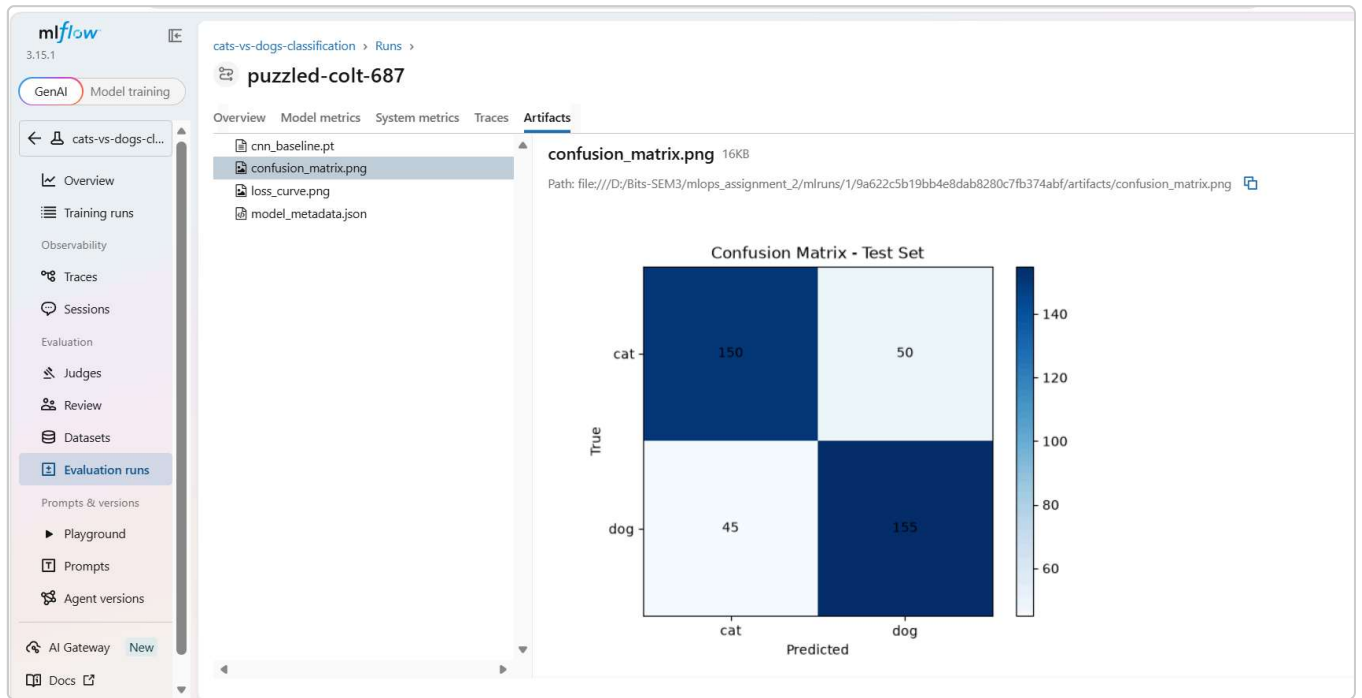
cnv_baseline.pt confusion_matrix.png loss_curve.png model_metadata.json

loss_curve.png 28.71KB

Path: file:///D:/Bits-SEM3/mlops_assignment_2/mlruns/1/9a622c5b19bb4e8dab8280c7fb374abf/artifacts/loss_curve.png

Training vs Validation Loss

epoch	train_loss	val_loss
0	0.65	0.65
1	0.60	0.60
2	0.58	0.58
3	0.55	0.55
4	0.52	0.52
5	0.50	0.50
6	0.48	0.48
7	0.47	0.47
8	0.46	0.46
9	0.45	0.45
10	0.44	0.44



M2: Model Packaging & Containerization 10M

Objective: Package the trained model into a reproducible, containerized service.

1. Inference Service

FastAPI (src/api/main.py) with GET /health and POST /predict (multipart image upload → label + class probabilities). Model loaded once at startup via a lifespan context manager.

```
curl http://localhost:8000/health
curl -X POST -F "file=@data/processed/test/dog/dog_00000.jpg" http://localhost:8000/predict
```

```
D:\Bits-SEM3\mlops_assignment_2>curl http://localhost:8000/health
{"status":"healthy","version":"1.1"}
D:\Bits-SEM3\mlops_assignment_2>curl -X POST -F "file=@data/processed/test/dog/dog_00000.jpg" http://localhost:8000/predict
{"label":"dog","probability":0.6634349226951599,"class_probabilities":{"cat":0.3365650773048401,"dog":0.6634349226951599}}
D:\Bits-SEM3\mlops_assignment_2>
```

2. Environment Specification

requirements.txt (full dev/training deps) and a leaner docker/requirements.txt (inference-only), both with pinned versions for every key ML library. PyTorch/torchvision installed from the CPU wheel index separately, since PyPI's default index only hosts CUDA builds.

3. Containerization

docker/Dockerfile : python:3.11-slim base, CPU torch/ torchvision installed as its own layer before app code, HEALTHCHECK polling /health every 30s.

```
docker build -f docker/Dockerfile -t cats-dogs-api:latest .
docker run -d --name cats-dogs-api -p 8000:8000 cats-dogs-api:latest
curl http://localhost:8000/health
```

docker build completing successfully, the container starting, and a healthy response confirming the fresh build works:


```

(venv) D:\Bits-SEM3\mlops_assignment_2>docker build -f docker/Dockerfile -t cats-dogs-api:latest .
[+] Building 351.8s (13/13) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 855B
=> [internal] load metadata for docker.io/library/python:3.11-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 143B
=> [internal] load build context
=> => transferring context: 10.01kB
=> [1/7] FROM docker.io/library/python:3.11-slim@sha256:1042b61448fef4b9a92d16a8c7eb4996d027568ce64792a7877fd88511e0af7c6
=> => resolve docker.io/library/python:3.11-slim@sha256:1042b61448fef4b9a92d16a8c7eb4996d027568ce64792a7877fd88511e0af7c6
=> => sha256:6310eb16bf4251731feab01e8f633bf5e2d75a657ccad97f420b1f83cce457be 28.83MB / 29.79MB
=> => extracting sha256:6310eb16bf4251731feab01e8f633bf5e2d75a657ccad97f420b1f83cce457be 345.6s
=> => extracting sha256:87e1b7cce023772f3a18c08fa0e287c519b474afbfcdd87ed023faad934cadb9 8.4s
=> => extracting sha256:c86306e32cd038ae40a4d1b8258a046618be19d6ef5280bb62a89960fecbf796 0.6s
=> => extracting sha256:a1457809edaf2df831b7865bfcc6d174e32bdeec920d1a66e11cab6269d7fa 7.8s
=> [2/7] WORKDIR /app
=> [3/7] RUN pip install --no-cache-dir --index-url https://download.pytorch.org/whl/cpu torch==2.13.0 torchvision==0.28.0
=> [4/7] COPY docker/requirements.txt .
=> [5/7] RUN pip install --no-cache-dir -r requirements.txt
=> [6/7] COPY src/ src/
=> [7/7] COPY models/cnn_baseline.pt models/cnn_baseline.pt
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:5a248f43c50fd92edc1b557357af25f1df2e7c5e5b59760a9724c429df51d33c 106.4s
=> => exporting config sha256:305636455986e7c1lace4ceea6cae1ae52238fa1b9f96c9359b60e18ba79e85b 0.0s
=> => exporting attestation manifest sha256:1a5b7fff54b3aaf48153bb2259468f37e8855327403be163bca6dafff4add54b1 0.1s
=> => exporting manifest list sha256:3f6ad1d53908337a74754b1b6f0b11b6782a08b905fe2f1a5065f62cec31cfba 0.0s
=> => naming to docker.io/library/cats-dogs-api:latest 0.2s
=> => unpacking to docker.io/library/cats-dogs-api:latest 65.8s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/q3foni0oxgh3etxj9lg5palaj

(venv) D:\Bits-SEM3\mlops_assignment_2>docker run -d --name cats-dogs-api -p 8000:8000 cats-dogs-api:latest
55ba4cda7030c9bcbad01dcfe3ae83c57c8f8ab9431a99323025e825e0e85f5a

(venv) D:\Bits-SEM3\mlops_assignment_2>curl http://localhost:8000/health
{"status": "healthy", "version": "1.1"}
(venv) D:\Bits-SEM3\mlops_assignment_2>

```


M3: CI Pipeline for Build, Test & Image Creation 10M

Objective: Implement Continuous Integration to automatically test, package, and build container images.

1. Automated Testing

12 pytest unit tests: 8 in tests/test_preprocess.py (is_valid_image() , split_paths()), 4 in tests/test_inference.py (predict() against synthetic images).

```
python -m pytest tests/ -v
```

```
(venv) D:\Bits-SEM3\mlops_assignment_2>python -m pytest tests/ -v
===== test session starts =====
platform win32 -- Python 3.11.9, pytest-8.4.2, pluggy-1.6.0 -- D:\Bits-SEM3\mlops_assignment_2\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: D:\Bits-SEM3\mlops_assignment_2
plugins: anyio-4.14.2, hydra-core-1.3.5
collected 12 items

tests/test_inference.py::test_predict_returns_expected_structure PASSED [ 8%]
tests/test_inference.py::test_predict_class_probabilities_sum_to_one PASSED [ 16%]
tests/test_inference.py::test_predict_handles_non_square_and_non_224_images PASSED [ 25%]
tests/test_inference.py::test_predict_handles_grayscale_input PASSED [ 33%]
tests/test_preprocess.py::test_is_valid_image_accepts_real_image PASSED [ 41%]
tests/test_preprocess.py::test_is_valid_image_rejects_corrupt_file PASSED [ 50%]
tests/test_preprocess.py::test_is_valid_image_rejects_empty_file PASSED [ 58%]
tests/test_preprocess.py::test_is_valid_image_rejects_missing_file PASSED [ 66%]
tests/test_preprocess.py::test_split_paths_respects_ratios PASSED [ 75%]
tests/test_preprocess.py::test_split_paths_no_overlap_and_covers_all PASSED [ 83%]
tests/test_preprocess.py::test_split_paths_is_deterministic_given_seed PASSED [ 91%]
tests/test_preprocess.py::test_split_paths_different_seeds_can_differ PASSED [100%]

===== 12 passed in 9.23s =====
```

2. CI Setup - GitHub Actions

.github/workflows/ci.yml , triggered on push/PR to main . test job: checkout → Python 3.11 → install deps → python -m pytest .

3. Artifact Publishing

build-and-push job (depends on test passing): builds the Docker image, **smoke-tests it locally first** (runs the container, polls /health , fails with logs if unhealthy), then pushes to **GitHub Container Registry** (ghcr.io/madihafathima/bits_mlops_assignment_2) using the automatically-provided GITHUB_TOKEN .

Workflow run history (the red entry is a deliberate negative test, covered under M4 below - all other runs on main are green):

⌵

MadihaFathima / bits_mlops_assignment_2

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<> Code

🔒 Issues

🔗 Pull requests

👤 Agents

🔄 Actions

📁 Projects

📖 Wiki

🛡 Security and quality

📊 Insights

⚙ Settings

Actions

New workflow

All workflows

CI

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

All workflows

Showing runs from all workflows

12 workflow runs

Workflow

Event

Status

Branch

Actor

🟢 Add README with setup and usage instructions

CI #12: Commit 7e94159 pushed by MadihaFathima

main

Today at 1:36 PM

6m 10s

...

🟢 Revert smoke test port; add post-deployment eval script and M4 negati...

CI #11: Commit 1641bcc pushed by MadihaFathima

main

Today at 1:01 PM

6m 51s

...

🔴 TEMP: deliberately break smoke test port to verify pipeline fails closed

CI #10: Commit 9615cfb pushed by MadihaFathima

main

Today at 12:45 PM

9m 18s

...

🟢 Add request logging and Prometheus metrics for M5 monitoring

CI #9: Commit f84ca1b pushed by MadihaFathima

main

Today at 12:18 PM

6m 44s

...

🟢 Increase rollout status timeout to 600s for slow multi-replica startup

CI #8: Commit fe444e1 pushed by MadihaFathima

main

Aug 26, 11:40 PM GMT+5:30

13m 10s

...

Job breakdown for a successful run - test, build-and-push, and deploy all succeeded:

< CI

🟢 Add README with setup and usage instructions #12

Summary

All jobs

test

build-and-push

deploy

Run details

Usage

Workflow file

Triggered via push 3 hours ago

MadihaFathima pushed

7e94159

main

Status

Success

Total duration

6m 10s

Artifacts

-

ci.yml

on: push

🟢 test

1m 21s

🟢 build-and-push

1m 25s

🟢 deploy

3m 16s

Published image in GitHub Container Registry:

⌵

MadihaFathima / bits_mlops_assignment_2

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<> Code

🔒 Issues

🔗 Pull requests

👤 Agents

🔄 Actions

📁 Projects

📖 Wiki

🛡 Security and quality

📊 Insights

⚙ Settings

bits_mlops_assignment_2

7e941594901d0e3e59404db887cc614a61e37378

Public

Latest

Install from the command line

Learn more about packages

\$ docker pull ghcr.io/madihafathima/bits_mlops_assignment_2:7e941594901d0e3e59404db887cc614a61e37378

Recent tagged image versions

latest

7e941594901d0e3e59404db887cc614a61e37378

Published about 4 hours ago · Digest

...

1641bcc5017224b00a074949a5267db3065e4c29

Published about 4 hours ago · Digest

...

9615cfb1cc8067db02a15456891ee56568242b50

Published about 4 hours ago · Digest

...

f84ca1ba88c034b730b2729ce34a5d00257d8cc3

Published about 4 hours ago · Digest

...

Details

MadihaFathima

bits_mlops_assignment_2

0 stars

Last published

4 hours ago

Issues

0

Total downloads

8

Contributors

1

MadihaFathima

Open an issue

Package settings

M4: CD Pipeline & Deployment 10M

Objective: Implement Continuous Deployment of the containerized model to a target environment.

1. Deployment Target

Local Kubernetes cluster (Docker Desktop's built-in Kubernetes). `k8s/deployment.yaml` (2 replicas, resource requests/limits, startup/ readiness/liveness probes on `/health`) and `k8s/service.yaml` (`LoadBalancer` , port 8080).

2. CD / GitOps Flow

A `deploy` job in the same GitHub Actions workflow, running on a **self-hosted runner** registered on the deployment machine (required since GitHub's cloud runners cannot reach a local cluster). Gated to push events on `main` only. Applies the manifests, then `kubectl set image` to the exact commit's image tag - guaranteeing a real rollout every time - and waits for rollout completion.

```
kubectl apply -f k8s/deployment.yaml
kubectl apply -f k8s/service.yaml
kubectl rollout status deployment/cats-dogs-api
curl http://localhost:8080/health
```

`kubectl get pods` (2/2 Running) and `kubectl get svc` (LoadBalancer with an external IP, on port 8080):

```
(venv) D:\Bits-SEM3\mlops_assignment_2>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
cats-dogs-api-774dc8bb4-9ngng       1/1     Running   0           3h33m
cats-dogs-api-774dc8bb4-g27mm       1/1     Running   0           3h31m
heart-disease-api-b9ccfc57f-ghphm   1/1     Running   9 (5h3m ago)  48d
heart-disease-api-b9ccfc57f-khkv5   1/1     Running   6 (5h3m ago)  48d

(venv) D:\Bits-SEM3\mlops_assignment_2>kubectl get svc
NAME                                TYPE           CLUSTER-IP      EXTERNAL-IP    PORT(S)          AGE
cats-dogs-api-service              LoadBalancer   10.96.132.251    172.18.0.5     8080:32526/TCP   6d16h
heart-disease-api-service          LoadBalancer   10.96.186.242    172.18.0.3     80:31642/TCP     48d
kubernetes                         ClusterIP       10.96.0.1        <none>         443/TCP          48d
```

The self-hosted runner listening for and processing deploy jobs (including one intentional failure from the negative test below):

```
D:\actions-runner>run.cmd
1 file(s) copied.

✓ Connected to GitHub

Current runner version: '2.336.0'
2026-08-30 06:43:31Z: Listening for Jobs

✓ Connected to GitHub

Current runner version: '2.336.0'
2026-08-30 06:44:04Z: Listening for Jobs
2026-08-30 06:51:37Z: Running job: deploy
2026-08-30 06:55:27Z: Job deploy completed with result: Succeeded
2026-08-30 07:18:46Z: Running job: deploy
2026-08-30 07:25:04Z: Job deploy completed with result: Failed
2026-08-30 07:34:07Z: Running job: deploy
2026-08-30 07:38:11Z: Job deploy completed with result: Succeeded
2026-08-30 08:09:46Z: Running job: deploy
2026-08-30 08:13:02Z: Job deploy completed with result: Succeeded
```

Full pipeline - test, build-and-push, and deploy all succeeded:

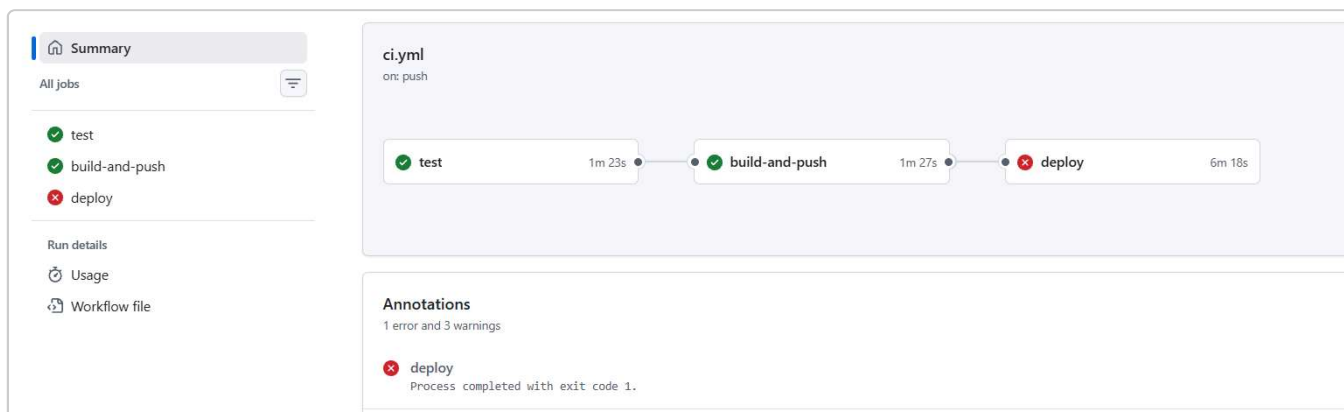
The screenshot shows the GitHub Actions interface for a workflow named 'Add README with setup and usage instructions #12'. The workflow is triggered by a push to the 'main' branch. The status is 'Success', and the total duration is '6m 10s'. The workflow consists of three jobs: 'test' (1m 21s), 'build-and-push' (1m 25s), and 'deploy' (3m 16s). All jobs are marked as successful with green checkmarks. The left sidebar shows the 'Summary' tab selected, with links for 'All jobs', 'Run details', 'Usage', and 'Workflow file'.

3. Smoke Tests / Health Check

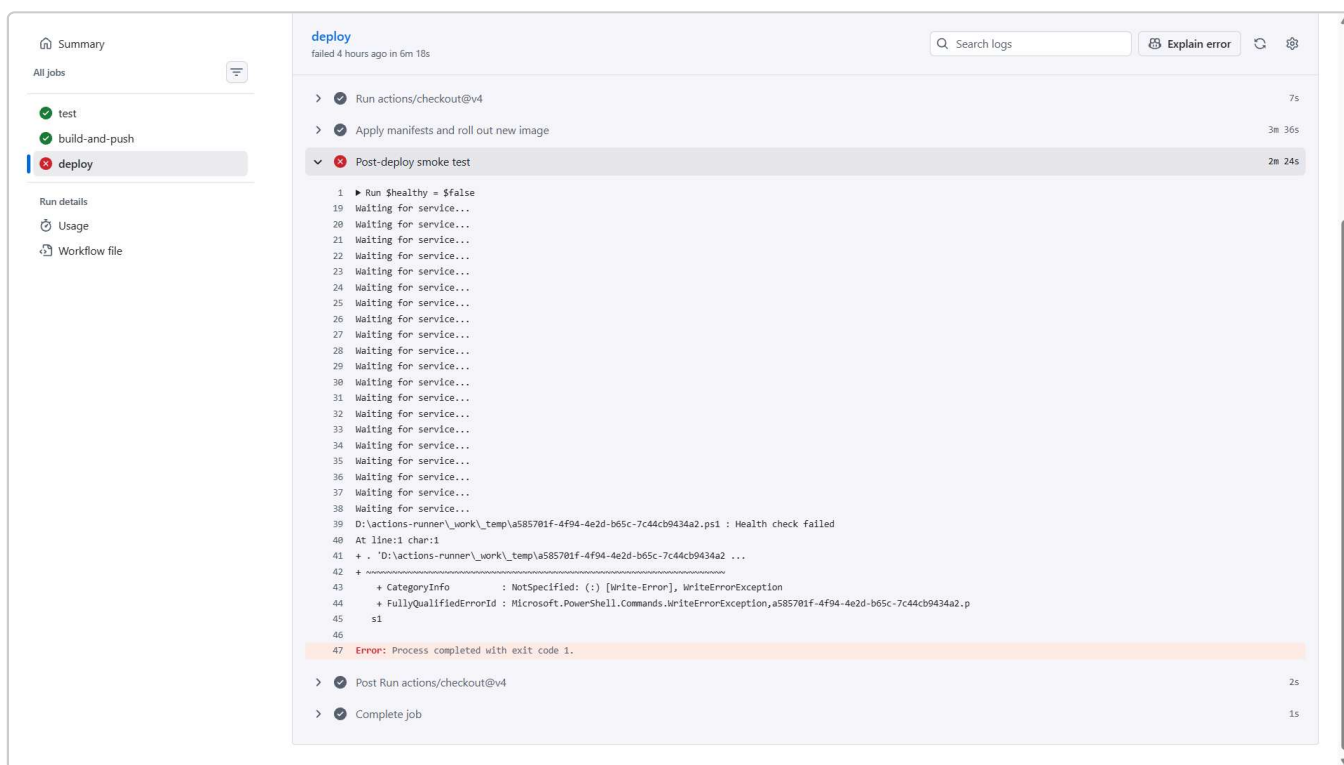
Post-deploy smoke test: polls `/health` (up to 20 retries), then a real `/predict` call using a fixture image committed to git. **Fails the pipeline** if either check fails.

Verified with a deliberate negative test: the smoke test was temporarily pointed at a wrong port. The real rollout still succeeded (proving the failure is isolated to the smoke test itself), but the smoke test step correctly failed, which correctly failed the whole `deploy` job and pipeline run - while the live service remained completely unaffected throughout. Reverted immediately after confirming.

The negative test run: test and build-and-push succeeded normally, but deploy failed - proving the pipeline fails closed:



Detailed log - the smoke test correctly detected the failure and exited with a non-zero code, failing the job:



M5: Monitoring, Logs & Final Submission 10M

Objective: Monitor the deployed model and submit a consolidated package of all artifacts.

1. Basic Monitoring & Logging

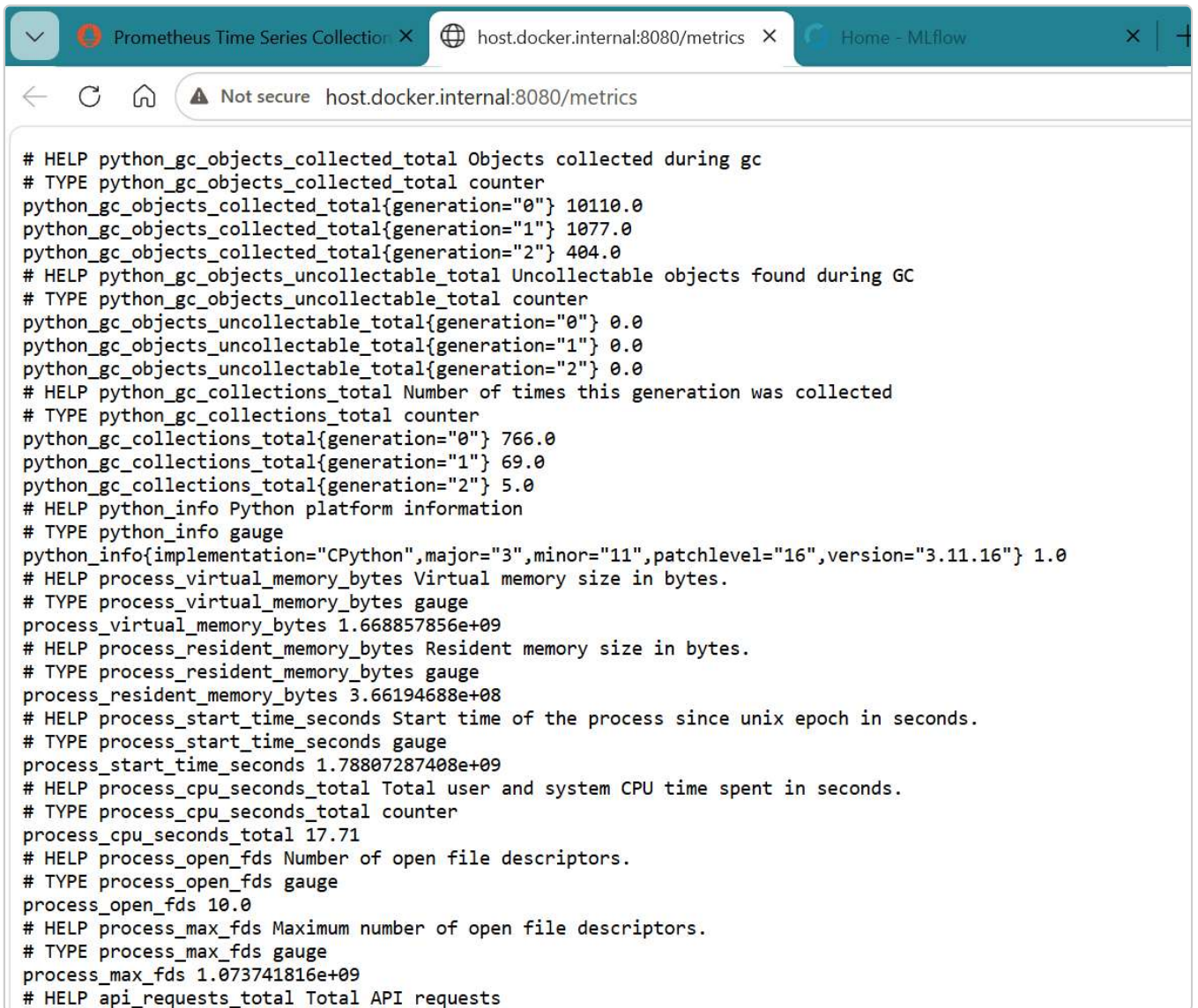
Every request logged (method, path, status code, latency in ms) via Python's `logging` module - metadata only, no request/response bodies. Prometheus- format metrics exposed at `GET /metrics` : `api_requests_total` (by method/path/status), `api_request_latency_seconds` (histogram, by path), `predictions_total` (by predicted label).

```
curl http://localhost:8080/metrics
docker run -d --name prometheus-cats-dogs -p 9090:9090 \
  -v "<repo-path>\monitoring\prometheus.yml:/etc/prometheus/prometheus.yml" prom/prometheus
```

The Prometheus container itself, running standalone and scraping the deployed service:

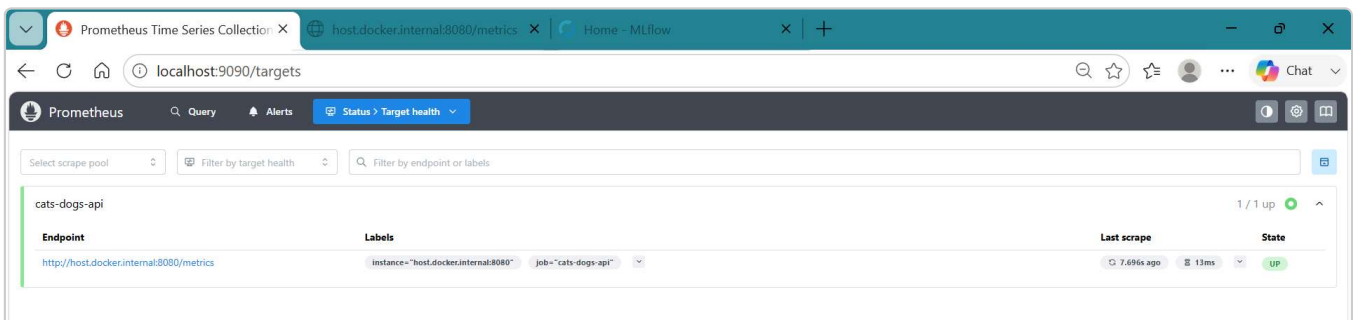
```
D:\Bits-SEM3\mlops_assignment_2>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
d77085691829   prom/prometheus  "/bin/prometheus --c..."  5 hours ago   Up 5 hours   0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp  prometheus-cats-dogs
D:\Bits-SEM3\mlops_assignment_2>
```

`/metrics` output on the live deployed service, showing populated counters:



```
# HELP python_gc_objects_collected_total Objects collected during gc
# TYPE python_gc_objects_collected_total counter
python_gc_objects_collected_total{generation="0"} 10110.0
python_gc_objects_collected_total{generation="1"} 1077.0
python_gc_objects_collected_total{generation="2"} 404.0
# HELP python_gc_objects_uncollectable_total Uncollectable objects found during GC
# TYPE python_gc_objects_uncollectable_total counter
python_gc_objects_uncollectable_total{generation="0"} 0.0
python_gc_objects_uncollectable_total{generation="1"} 0.0
python_gc_objects_uncollectable_total{generation="2"} 0.0
# HELP python_gc_collections_total Number of times this generation was collected
# TYPE python_gc_collections_total counter
python_gc_collections_total{generation="0"} 766.0
python_gc_collections_total{generation="1"} 69.0
python_gc_collections_total{generation="2"} 5.0
# HELP python_info Python platform information
# TYPE python_info gauge
python_info{implementation="CPython",major="3",minor="11",patchlevel="16",version="3.11.16"} 1.0
# HELP process_virtual_memory_bytes Virtual memory size in bytes.
# TYPE process_virtual_memory_bytes gauge
process_virtual_memory_bytes 1.668857856e+09
# HELP process_resident_memory_bytes Resident memory size in bytes.
# TYPE process_resident_memory_bytes gauge
process_resident_memory_bytes 3.66194688e+08
# HELP process_start_time_seconds Start time of the process since unix epoch in seconds.
# TYPE process_start_time_seconds gauge
process_start_time_seconds 1.78807287408e+09
# HELP process_cpu_seconds_total Total user and system CPU time spent in seconds.
# TYPE process_cpu_seconds_total counter
process_cpu_seconds_total 17.71
# HELP process_open_fds Number of open file descriptors.
# TYPE process_open_fds gauge
process_open_fds 10.0
# HELP process_max_fds Maximum number of open file descriptors.
# TYPE process_max_fds gauge
process_max_fds 1.073741816e+09
# HELP api_requests_total Total API requests
```

Prometheus Target health - cats-dogs-api scraped successfully (1/1 up):



2. Model Performance Tracking (Post-Deployment)

scripts/post_deploy_eval.py sends a batch of real requests - test images with known true labels - to the live deployed /predict endpoint, compares predictions against ground truth, and reports accuracy. Report written to outputs/post_deploy_eval.json .

```
python scripts/post_deploy_eval.py
```


Running the script against the live deployed service - 30 real requests, each compared against its known true label:

```
(venv) D:\Bits-SEM3\mlops_assignment_2>python scripts/post_deploy_eval.py
Sending 30 real requests to http://localhost:8080/predict ...
cat_00009.jpg: true=cat pred=cat correct=True
dog_00085.jpg: true=dog pred=dog correct=True
dog_00017.jpg: true=dog pred=dog correct=True
cat_00085.jpg: true=cat pred=dog correct=False
dog_00000.jpg: true=dog pred=dog correct=True
dog_00096.jpg: true=dog pred=dog correct=True
dog_00198.jpg: true=dog pred=cat correct=False
dog_00187.jpg: true=dog pred=dog correct=True
cat_00087.jpg: true=cat pred=cat correct=True
dog_00080.jpg: true=dog pred=dog correct=True
cat_00097.jpg: true=cat pred=cat correct=True
dog_00179.jpg: true=dog pred=dog correct=True
cat_00027.jpg: true=cat pred=cat correct=True
dog_00022.jpg: true=dog pred=dog correct=True
cat_00040.jpg: true=cat pred=cat correct=True
dog_00001.jpg: true=dog pred=dog correct=True
dog_00041.jpg: true=dog pred=cat correct=False
cat_00137.jpg: true=cat pred=dog correct=False
dog_00143.jpg: true=dog pred=dog correct=True
dog_00062.jpg: true=dog pred=dog correct=True
cat_00034.jpg: true=cat pred=cat correct=True
cat_00143.jpg: true=cat pred=cat correct=True
cat_00013.jpg: true=cat pred=cat correct=True
dog_00152.jpg: true=dog pred=cat correct=False
cat_00104.jpg: true=cat pred=cat correct=True
dog_00111.jpg: true=dog pred=dog correct=True
cat_00022.jpg: true=cat pred=dog correct=False
cat_00068.jpg: true=cat pred=cat correct=True
cat_00196.jpg: true=cat pred=dog correct=False
cat_00086.jpg: true=cat pred=cat correct=True

Post-deployment accuracy: 0.7667 (23/30)
Full report written to outputs\post_deploy_eval.json
```

The full JSON report written to outputs/post_deploy_eval.json :

```
{
  "base_url": "http://localhost:8080",
  "n_requests": 30,
  "accuracy": 0.7666666666666667,
  "results": [
    {
      "file": "cat_00009.jpg",
      "true_label": "cat",
      "predicted_label": "cat",
      "probability": 0.6271777451038361,
      "correct": true
    },
    {
      "file": "dog_00085.jpg",
      "true_label": "dog",
      "predicted_label": "dog",
      "probability": 0.8881039619445801,
      "correct": true
    },
    {
      "file": "dog_00017.jpg",
      "true_label": "dog",
      "predicted_label": "dog",
      "probability": 0.6572423577308655,
      "correct": true
    },
    {
      "file": "cat_00085.jpg",
      "true_label": "cat",
      "predicted_label": "dog",
      "probability": 0.5688731670379639,
      "correct": false
    },
    {
      "file": "dog_00000.jpg",
      "true_label": "dog",
      "predicted_label": "dog",
      "probability": 0.6634349226951599,
      "correct": true
    },
    {
      "file": "dog_00096.jpg",
      "true_label": "dog",
      "predicted_label": "dog",
      "probability": 0.5665894746780396,
      "correct": true
    }
  ]
}
```

Pipeline Summary

Stage	Tool
Data versioning	Git + DVC
Model	PyTorch (SimpleCNN)
Experiment tracking	MLflow
Serving	FastAPI
Containerization	Docker
CI	GitHub Actions
CD	GitHub Actions (self-hosted runner) + Kubernetes
Monitoring	Python logging + Prometheus

Deliverables

Repository: https://github.com/MadihaFathima/bits_mlops_assignment_2

Demo video: https://wilpbitspilaniacin0-my.sharepoint.com/:v/g/personal/2024ad05394_wilp_bits-pilani_ac_in/IQCr6euuEanUSqQLxxqPhKBaAUOIAGzVEyK1JlfomLOphbs?e=BnI9jd - screen recording demonstrating the complete workflow.